Steel Plate Fault Analysis

A Six Sigma DMAIC Project on 1,941 Faulty Steel Plates

Personal project using a public dataset

Tools: jamovi, Google Sheets, draw.io

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October 2026

Data source: Buscema, Terzi & Tastle (2010), Steel Plates Faults, UCI Machine Learning Repository (CC BY 4.0).

Abstract

Steel plate makers need to know which faults to tackle first. This project applies the DMAIC steps to 1,941 faulty plates from a public dataset with seven fault types, using a Pareto chart, Welch's ANOVA with Games-Howell comparisons, a chi-square test and a fishbone diagram. Other_Faults, Bumps and K_Scatch make up about 75% of faults, and K_Scatch occurs almost only on thin (40 mm) A400 plates (99.7% on A400). The data show patterns, not causes, so the project ends with targeted trials and a control chart shown only as a method demonstration.

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1. Introduction (Define)

1.1 Problem statement

Steel plates are inspected for surface and shape faults before they go to customers. A plate with a fault may be downgraded, repaired or scrapped, so every fault costs time and material. The inspection records list seven fault types, and each type may have different causes. If effort is spread evenly across all seven, the biggest problems get no more attention than the smallest.

The question for this project is simple: **which faults happen most, and do they follow any pattern in plate thickness, steel type or conveyor position that points to where to investigate?**

1.2 Objective

The objective is to use the DMAIC method (Define, Measure, Analyze, Improve, Control) on a public dataset to do three things:

- rank the fault types by how often they occur, so the vital few can be separated from the rest;
- test whether the fault types differ in plate thickness and steel type, and describe how they differ in conveyor length value;
- turn the findings into a short list of trials and a monitoring plan that a plant team could follow.

1.3 Scope

In scope: the 1,941 faulty plate records in the UCI Steel Plates Faults dataset, the seven fault types, plate thickness, steel type (A300 or A400) and the conveyor length value. **Out of scope:** healthy plates, cost data, any real production line, and image-level analysis of the fault shapes. The project is a learning exercise built on public data. It does not describe a real company, and no improvement result is claimed.

1.4 How DMAIC is used here

Phase	Question asked	Tool used	Where
Define	What is the problem and the scope?	Problem statement	Section 1
Measure	Which faults are most frequent?	Pareto chart	Section 3
Analyze	Do faults differ by thickness, steel type or conveyor value?	Box plot, Welch's ANOVA, Games-Howell, chi-square, fishbone	Section 4
Improve	What should be tried next?	Recommendations table	Section 5
Control	How would results be tracked?	p-chart (method demonstration)	Section 6

Table 1. The five DMAIC phases, the question each answers, and the tool used.

2. Data and Methods

2.1 The dataset

The data come from the Steel Plates Faults dataset in the UCI Machine Learning Repository, donated in 2010 and created by Buscema, Terzi and Tastle. It holds **1,941 records**, one per faulty plate, with **27 measurements** taken from the fault images and plate details, and **7 fault types**: Pastry, Z_Scratch, K_Scatch, Stains, Dirtiness, Bumps and Other_Faults. The repository reports no missing values, and this was confirmed when the files were loaded. The dataset is released under the CC BY 4.0 licence.

The original file stores the fault type as seven yes/no columns. A check showed that every record has exactly one “yes”, so the seven columns were combined into one column called Fault_Type. This gave a clean file with 27 measurements plus Fault_Type, 1,941 rows and no duplicate rows. Three files are kept in the project: the original CSV, the clean CSV, and the Pareto table used for Figure 1.

2.2 Variables used

Variable	Type	Why it was used
Fault_Type	Category, 7 levels	The outcome: which fault the plate has
Steel_Plate_Thickness	Number (mm)	Compared across fault types
TypeOfSteel_A300 / A400	Yes/no pair	Which steel type the plate is made of
Length_of_Conveyer	Number	A position-related clue; the repository does not explain it further, so it is described only, not tested

Table 2. Variables used in the analysis. The other 23 measurements (fault size, luminosity and shape indices) were not needed for the questions asked.

2.3 Tools

Tool	Used for
Google Sheets	Counting faults and drawing the Pareto chart
jamovi (version 2.7, built on R 4.5)	Descriptives, box plot, Welch’s ANOVA, Games-Howell test, contingency table and chi-square test
draw.io	Fishbone diagram
Python (matplotlib)	p-chart for the Control phase, and cross-checks of the counts and percentages in this report

Table 3. Tools used and what each was used for.

2.4 Tests used and why

Pareto analysis. Faults are counted and sorted from largest to smallest, with a running percentage. It shows the few fault types that account for most of the problem.

Welch’s one-way ANOVA. This asks whether the average plate thickness differs between the seven fault types. Welch’s version was chosen over the standard ANOVA because the groups have very different spreads (the standard deviation is 2.2 mm for K_Scatch but 73.9 mm for Other_Faults) and very different sizes (55 to 673 plates). The standard test assumes similar spreads, so it would be unreliable here.

Games-Howell post-hoc test. ANOVA only says that *some* groups differ. Games-Howell compares every pair of fault types and does not assume equal spreads, so it shows *which* pairs differ. It also adjusts for the fact that 21 pairs are compared at once.

Contingency table and chi-square test. Steel type is a category, not a number, so it cannot go into an ANOVA. A contingency table counts plates for each combination of fault type and steel type, and the chi-square test checks whether the fault types are spread across the two steel types more unevenly than chance would explain.

Descriptive table for conveyor length value. The values take only a narrow range and their meaning is not documented, so they are summarised (count, mean, median, spread, minimum, maximum) rather than tested.

Fishbone diagram and p-chart. The fishbone organises possible causes by category. The p-chart is the standard chart for tracking a proportion over time. Here it is used only to show the method.

A significance level of 0.05 was used throughout. A p-value below .001 is reported as “< .001”, as jamovi does.

3. Measure

The Measure phase answers a counting question: how many faulty plates are there of each type? Table 4 gives the counts and Figure 1 shows them as a Pareto chart.

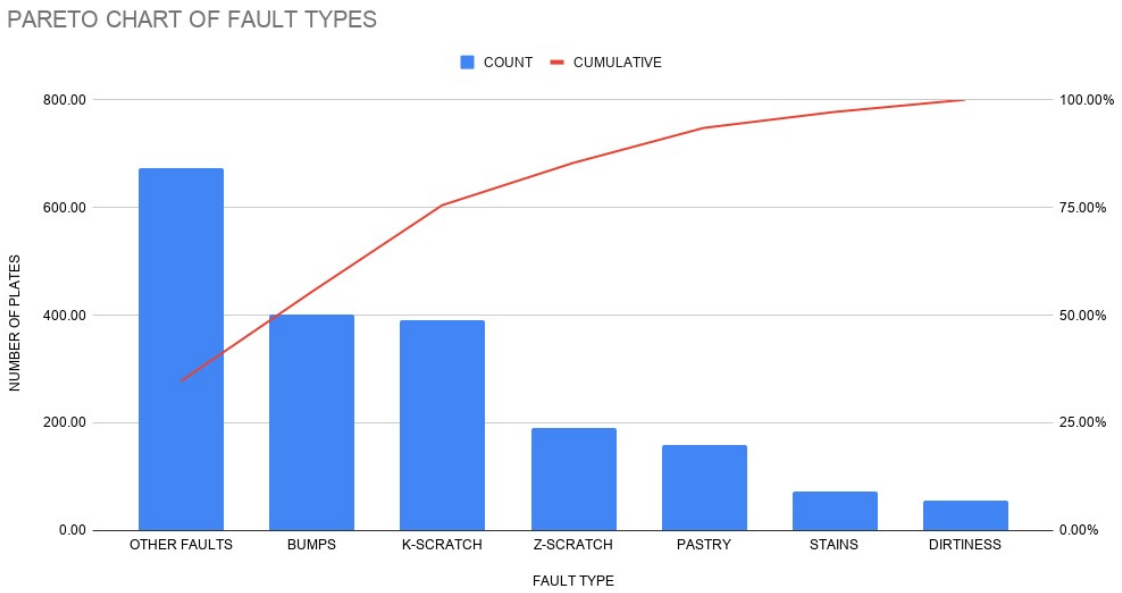


Figure 1. Pareto chart of the seven fault types (bars show the number of plates, the line shows the cumulative percentage).

What it shows: three fault types, Other_Faults, Bumps and K_Scatch, account for 1,466 of the 1,941 plates, which is about 75.5%.

Fault type	Plates	Share of all faults	Cumulative share
Other_Faults	673	34.7%	34.7%
Bumps	402	20.7%	55.4%
K_Scatch	391	20.1%	75.5%
Z_Scratch	190	9.8%	85.3%
Pastry	158	8.1%	93.5%
Stains	72	3.7%	97.2%
Dirtiness	55	2.8%	100.0%
Total	1,941	100.0%	

Table 4. Fault counts, share and cumulative share (the data behind Figure 1).

Two points are worth making. First, Other_Faults is the single largest group, but it is a catch-all label, so it cannot point to a specific cause until it is broken down into clearer categories. Second, Bumps and K_Scatch are the two largest *named* faults, and together they make up 40.9% of all faults. These two, plus Z_Scratch, are the faults the Analyze phase looks at most closely, because they showed the clearest patterns.

The percentages describe the share of *faulty* plates only. Because the dataset has no record of how many plates were made or inspected, it is not possible to say how often a plate is faulty, only

how the faults are divided among themselves.

4. Analyze

4.1 Plate thickness by fault type

Figure 2 and Table 5 compare the thickness of the plates for each fault type. Thickness is measured in millimetres, and the thinnest plate in the data is 40 mm.

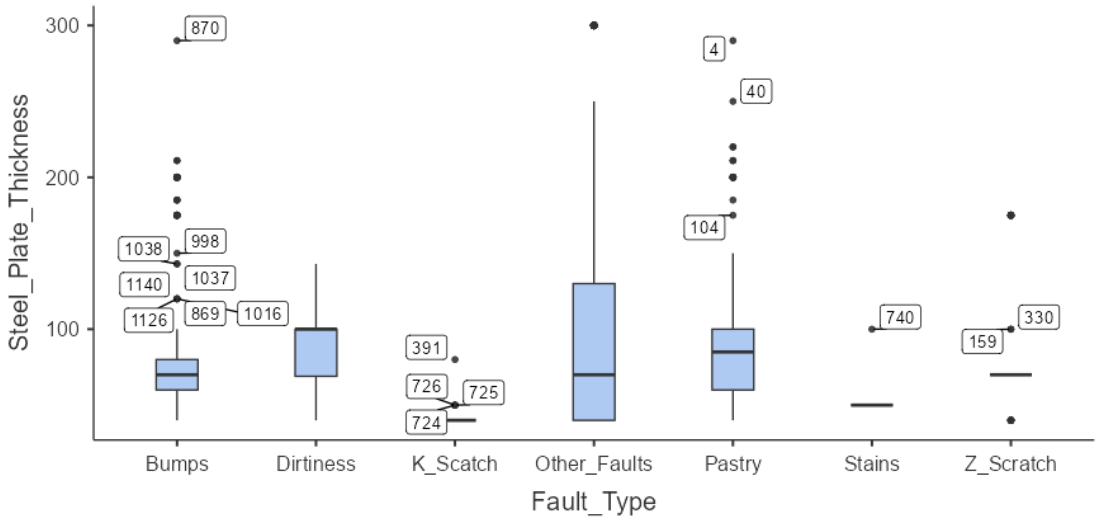


Figure 2. Box plot of plate thickness for each fault type (boxes show the middle half of the plates, the line is the median, and the dots are unusually thick plates; numbers are row labels from jamovi).

What it shows: K_Scatch plates are almost all the same thin thickness (40 mm), while Other_Faults and Pastry cover a very wide range of thicknesses.

Fault type	N	Mean	Median	SD	Min	Max
Bumps	402	76.9	70	38.0	40	290
Dirtiness	55	84.5	100	28.7	40	143
K_Scatch	391	40.2	40	2.20	40	80
Other_Faults	673	103	70	73.9	40	300
Pastry	158	92.9	85	51.6	40	290
Stains	72	50.7	50	5.89	50	100
Z_Scratch	190	74.6	70	24.7	40	175

Table 5. Plate thickness (mm) by fault type, from jamovi. N = number of plates, SD = standard deviation.

What it shows: K_Scatch has the lowest average thickness (40.2 mm) and by far the smallest spread (SD 2.2 mm); Stains is the next thinnest (50.7 mm).

4.2 Welch's ANOVA

Test	F	df1	df2	p
Steel_Plate_Thickness by Fault_Type	283	6	337	< .001

Table 6. Welch's one-way ANOVA for plate thickness across the seven fault types.

What it shows: average plate thickness is not the same across the fault types ($F = 283$, $p < .001$), so at least one fault type sits at a different thickness from the others. The next step finds which

ones.

4.3 Games-Howell comparisons

Table 7 lists the difference in mean thickness (row minus column) for each pair, with the p-value underneath.

	Dirtiness	K_Scatch	Other_Faults	Pastry	Stains	Z_Scratch
Bumps	-7.52 p = .586	36.8 p < .001	-25.7 p < .001	-15.97 p = .009	26.2 p < .001	2.35 p = .972
Dirtiness		44.3 p < .001	-18.1 p = .004	-8.45 p = .745	33.8 p < .001	9.88 p = .248
K_Scatch			-62.4 p < .001	-52.73 p < .001	-10.5 p < .001	-34.40 p < .001
Other_Faults				9.70 p = .455	51.9 p < .001	28.02 p < .001
Pastry					42.2 p < .001	18.33 p = .001
Stains						-23.88 p < .001

Table 7. Games-Howell post-hoc test: mean difference in thickness (mm), row minus column, and p-value.

What it shows: K_Scatch differs significantly from every other fault type (all p < .001), and Stains also differs from all the others, while Bumps and Z_Scratch (p = .972) and Pastry and Other_Faults (p = .455) cannot be told apart on thickness.

Because K_Scatch plates sit about 10 to 62 mm thinner than every other group, thickness is the clearest numeric signal in the dataset. A word of caution is needed, though: the thickness values are skewed, with a long tail of thick plates in several groups (see the dots in Figure 2). Welch's test copes with unequal spreads, but not with strong skew, so the p-values should be read as a clear signal of a difference rather than as exact probabilities.

4.4 Fault type by steel type

Table 8 shows how many plates of each fault type are made of steel A300 and A400 (in the jamovi output, the column "0" under TypeOfSteel_A300 means A400).

Fault type	A400 (plates)	A300 (plates)	Total	A300 share of row
Bumps	123	279	402	69.4%
Dirtiness	46	9	55	16.4%
K_Scatch	390	1	391	0.3%
Other_Faults	407	266	673	39.5%
Pastry	109	49	158	31.0%
Stains	71	1	72	1.4%
Z_Scratch	18	172	190	90.5%
Total	1,164	777	1,941	40.0%

Table 8. Fault type by steel type (contingency table from jamovi). Chi-square = 667, df = 6, p < .001, N = 1,941.

What it shows: K_Scatch is almost entirely on A400 plates (390 of 391, which is 99.7%), while Z_Scratch (90.5%) and Bumps (69.4%) are mostly on A300 plates.

The chi-square test confirms that fault type and steel type are strongly linked (chi-square = 667, df = 6, p < .001). As a measure of how strong the link is, Cramer's V, calculated from these numbers, is about 0.59, which is a large effect.

Because A400 plates are 60% of the faulty plates, some K_Scatch on A400 is expected by chance. The point is how far above that K_Scatch lies. Table 9 turns the table around and shows what share of each steel type's faults is K_Scatch.

Steel type	Faulty plates	K_Scatch plates	K_Scatch share of faults
A400	1,164	390	33.5%
A300	777	1	0.1%

Table 9. K_Scatch share of faults within each steel type (calculated from the clean CSV).

What it shows: about one in three faults on A400 plates is K_Scatch, compared with about one in a thousand on A300 plates.

Thickness adds a second layer. Of the 391 K_Scatch plates, 387 (99.0%) are 40 mm thick, and all 387 are A400 (the one A300 K_Scatch plate is not 40 mm). Among all 710 plates that are 40 mm thick, 54.5% have K_Scatch. One more finding limits what can be concluded: every 40 mm plate in the data is A400 (710 of 710), so thickness and steel type are tangled together. The data can say that K_Scatch is linked to thin A400 plates, but it cannot say whether thinness, the steel type, or the way that product is processed is responsible.

4.5 Conveyor length value

Table 10 summarises the Length_of_Conveyer value for each fault type. The dataset documentation does not explain what this number measures, so it is treated as a clue about where on the line a fault was found or how the plate was handled, and no test was run on it.

Fault type	N	Mean	Median	SD	Min	Max
Bumps	402	1523	1624	162	1227	1794
Dirtiness	55	1415	1364	117	1352	1696
K_Scatch	391	1384	1362	50.3	1336	1687
Other_Faults	673	1485	1372	154	1227	1715
Pastry	158	1554	1648	155	1333	1710
Stains	72	1358	1358	1.49	1354	1362
Z_Scratch	190	1358	1356	25.7	1348	1708

Table 10. Conveyor length value by fault type, from jamovi.

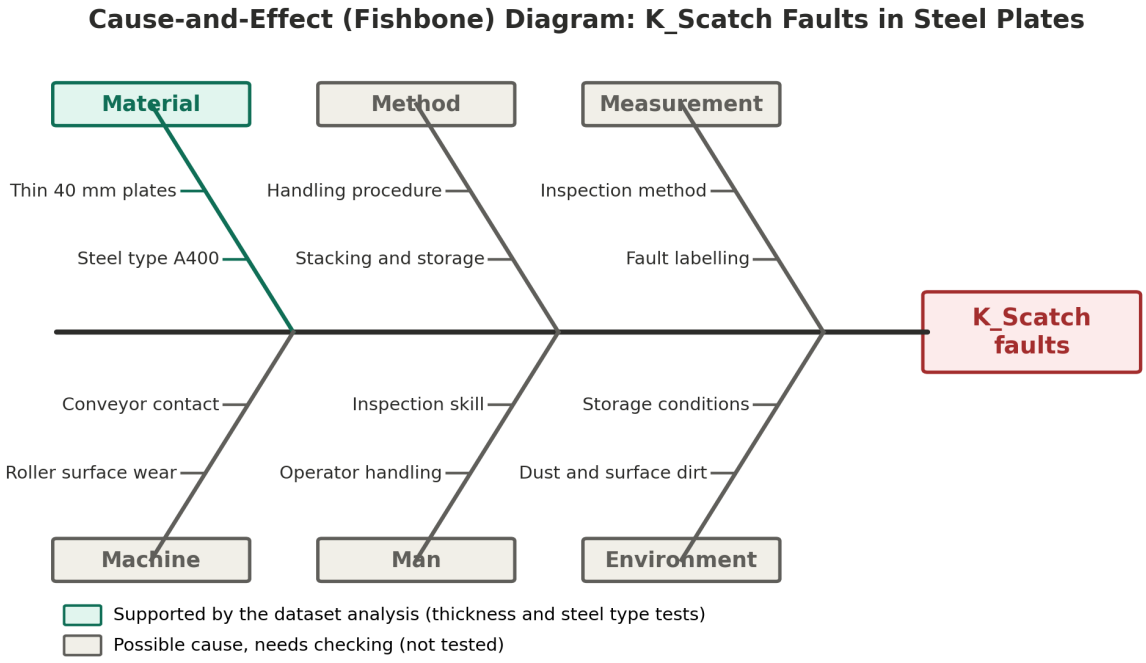
What it shows: Stains, Z_Scratch and K_Scatch cluster tightly around 1,360, whereas Bumps and Pastry have medians above 1,600 and a much wider spread.

Two groups stand out. Stains are almost always at the same value (SD 1.49), and Z_Scratch is nearly as tight (SD 25.7) apart from a few high values. Bumps and Pastry are different: their medians (1,624 and 1,648) are well above their means (1,523 and 1,554), which hints at a

separate group of plates with low values in those fault types. These are patterns to check in a real plant, not results.

4.6 Fishbone diagram

Figure 3 organises possible causes of K_Scatch into six standard categories. K_Scatch was chosen because it has the strongest and cleanest pattern in the data and it is the third biggest fault.



Source: UCI Steel Plates Faults dataset; analysis in jamovi. Grey branches are hypotheses from general process knowledge.

Figure 3. Fishbone diagram for K_Scatch faults. The green branch (Material) is supported by the thickness and steel type tests; the grey branches are possible causes from general process knowledge and were not tested.

What it shows: only the Material branch (thin 40 mm plates, steel type A400) is backed by data in this project; the Machine, Method, Man, Measurement and Environment branches are ideas to check, not findings.

The “conveyor contact” and “roller surface wear” ideas under Machine are placed on the diagram because the conveyor values for K_Scatch are tightly grouped, which would fit a fault that happens at a particular place on the line. This is a reasonable thing to look into, but the dataset cannot confirm it.

4.7 Summary of the Analyze findings

- K_Scatch plates are thin (mean 40.2 mm) and differ from every other fault type in thickness (Games-Howell, all $p < .001$).
- K_Scatch occurs almost only on A400 steel (99.7%); Z_Scratch (90.5%) and Bumps (69.4%) occur mostly on A300 steel.
- Fault type and steel type are strongly linked (chi-square = 667, $p < .001$, Cramer’s V about 0.59).

- Thickness and steel type are tangled in this data, since all 40 mm plates are A400, so their separate effects cannot be pulled apart.
- Conveyor length values cluster tightly for Stains, Z_Scratch and K_Scatch, which is a clue to investigate, not a conclusion.

5. Improve

Improvement actions in this project are **recommendations for trials**. None has been carried out, so no result is claimed. Each row of Table 11 links a finding to a trial and says how success would be checked. No target percentages are set, because the data has no baseline rate (see Section 7).

No.	Fault (share)	Evidence from this project	Recommended trial or action	How to check
1	K_Scatch (20.1%)	99.7% on A400; 99.0% are 40 mm plates; conveyor values tightly grouped (SD 50.3)	Trial on 40 mm A400 plates: inspect conveyor and roller contact points and plate handling on that route; compare plates that touch those points with plates that do not	K_Scatch share of faults per batch (p-chart); count of K_Scatch per 100 plates once total output is recorded
2	Bumps (20.7%)	69.4% on A300; median thickness 70 mm; conveyor median 1,624	Review handling, stacking and storage of A300 plates around 70 mm; look at whether the high conveyor values (above 1,600) match a particular station	Bump share among A300 faults; compare before and after any handling change
3	Z_Scratch (9.8%)	90.5% on A300; median thickness 70 mm; conveyor values near 1,358	Check contact and handling of A300 plates at the position that matches this conveyor value; compare with Bumps findings, since both sit on A300	Z_Scratch share among A300 faults
4	Other_Faults (34.7%)	Largest group but no single pattern: 39.5% on A300, thickness 40 to 300 mm	Re-label a sample of these plates into specific fault categories so they can be analysed; add new category names where needed	Share of faults still labelled Other_Faults
5	Pastry, Stains, Dirtiness (14.7% together)	Small counts; Stains almost always at one conveyor value (SD 1.49) and 98.6% on A400	Monitor only for now; revisit if their share grows. Check storage and cleanliness for Stains and Dirtiness	Monthly count and share
6	All faults	No time, plate ID, total output or specification limits in the data	Start recording date and time, plate ID, shift, number of plates inspected and the specification limits for each plate	Data fields complete for every inspected plate

Table 11. Improve: recommendations, numbered in order of priority, with the evidence behind each and how it would be checked. All are proposed trials, not completed actions.

The order of priority follows two things: how many plates are affected (Figure 1), and how clear the pattern in the data is (Section 4). K_Scatch comes first because it is the third largest fault and also has by far the clearest pattern. Other_Faults is listed fourth even though it is the biggest group, because without clearer labels there is nothing specific to trial.

A sensible way to run trial 1 would be a small, planned comparison: change one thing on a defined set of 40 mm A400 plates, leave a comparable set unchanged, and record K_Scatch outcomes for both. Changing several things at once would make it impossible to tell which change worked.

6. Control

The Control phase is about keeping an improvement in place and noticing quickly if the process slips. The standard tool for tracking the share of items with a given fault is a **p-chart**. It plots the share for each batch, with a centre line at the overall average and two limits set three standard errors above and below it. A point outside the limits, or a long run of points on one side of the centre line, suggests that something has changed.

Important: Figure 4 is a method demonstration only. The dataset has no time column, and the rows are sorted by fault type, so the file order says nothing about production order. To show how the chart is built, the 1,941 plates were placed in random order and cut into 20 batches of 97 or 98 plates. The chart then plots the K_Scatch share among faulty plates in each batch. It cannot show whether any real process is stable or improving.

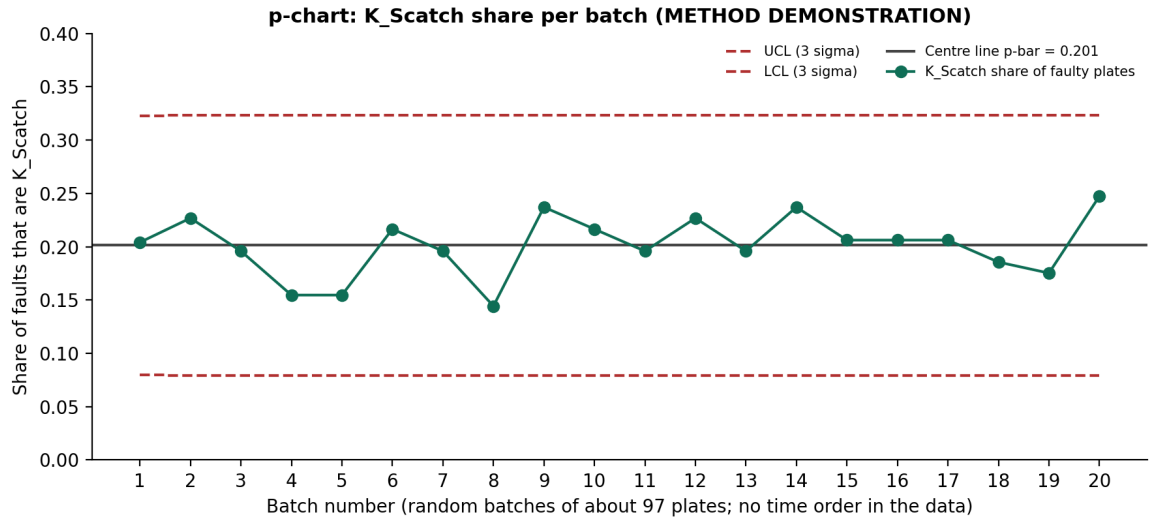


Figure 4. p-chart of the K_Scatch share of faulty plates in 20 random batches (method demonstration only; the batches are not time-ordered).

What it shows: the K_Scatch share varies between about 14% and 25% from batch to batch around a centre line of 20.1%, and every point lies inside the control limits (about 8% to 32%).

That result is expected. Random batches contain only chance variation, so they stay inside the limits. This is the baseline a real chart would start from. In a real plant, the same chart would be updated each week or shift, and a drop below the lower limit after a trial would be a sign that the trial changed something.

One more point on the choice of measure: because only faulty plates are in the data, the chart tracks the *share of faults that are K_Scatch*, not the *rate of K_Scatch per plate produced*. If total output were recorded (see Table 11, action 6), the chart could track the true fault rate, which is the better measure.

Item	Plan
What is plotted	K_Scatch share of faults per batch (later: K_Scatch per plates produced)
Batch	One shift or one week, using real time order
Limits	Centre line and 3-sigma limits from at least 20 batches before any trial starts
Signals to act on	A point outside a limit, or 8 points in a row on one side of the centre line
Reaction	Check the batch for changes in plate type, handling or equipment; record what was found
Review	Reset the limits only after a confirmed, lasting change

Table 12. How the control chart would be used in practice (a proposed plan, not an existing one).

7. Limitations

The findings should be read with the following limits in mind.

- **Public data.** The records come from a published research dataset, not from a plant I had access to. Nothing here describes any real company or line, and the results may not transfer to another process.
- **Only faulty plates.** With no healthy plates in the file, the fault *rate* cannot be calculated, only how the faults are divided among themselves. For example, “K_Scatch is common on A400” means K_Scatch is a large share of A400 faults, not that A400 plates are more likely to be faulty.
- **No time column.** Without dates or plate order, trends, shifts and real control chart signals cannot be examined. The p-chart in Section 6 is therefore a demonstration, not evidence.
- **No specification limits.** The dataset has no tolerances, so process capability measures such as Cp and Cpk cannot be worked out, and it is not possible to say which plates were out of specification.
- **Tangled variables.** All 40 mm plates are A400, so the effect of thickness cannot be separated from the effect of steel type.
- **Unclear conveyor variable.** The meaning of Length_of_Conveyer is not documented, so conclusions about conveyor position are only suggestions.
- **Statistical assumptions.** Thickness is skewed and the groups differ widely in size and spread. Welch’s test and Games-Howell handle unequal spreads but not strong skew, so p-values are best read as strong signs of a difference.
- **Association, not cause.** The tests show that faults and plate features go together. Proving a cause needs a controlled trial, which is what the Improve section proposes.
- **Labelling.** The fault labels were assigned by the dataset creators. Other_Faults is a catch-all, and some plates may have been labelled inconsistently.

8. Conclusion

Using the DMAIC steps on 1,941 faulty plates, this project found that three fault types, Other_Faults, Bumps and K_Scatch, account for about 75.5% of all faults. Among these, K_Scatch has the clearest pattern: it occurs almost only on thin (40 mm) plates of steel type A400 (99.7% on A400), and its thickness differs significantly from every other fault type. Bumps and Z_Scratch lean towards A300 plates around 70 mm thick (69.4% and 90.5% on A300).

The data cannot prove why this happens, and thickness and steel type cannot be separated because the 40 mm plates are all A400. For that reason the main output is not a claim of improvement but a practical plan: run targeted trials on 40 mm A400 plates and on A300 plates around 70 mm, relabel Other_Faults more precisely, and start recording time, plate ID, total output and specification limits. With those fields in place, a real control chart could replace the demonstration in Figure 4.

The project also shows a general lesson. A Pareto chart tells you where to look, a few well-chosen tests tell you what is different, and a fishbone diagram keeps untested ideas separate from supported findings. The last step is only possible when the right data are collected.

References

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Appendix: project files

File	Content
steel_plates_faults_original.csv	Original dataset: 27 measurements and 7 yes/no fault columns (1,941 rows)
steel_plates_faults_clean.csv	Cleaned file: 27 measurements and one Fault_Type column (1,941 rows)
PARETO_STEEL_PROJECT.csv	Fault counts and cumulative percentages used for Figure 1
jamovi project file and results PDF	Descriptives, box plot, Welch's ANOVA, Games-Howell, contingency table, chi-square and conveyor summary
Figures	Pareto chart, box plot, fishbone diagram, p-chart
This report (PDF)	Final project report

Table 13. Files included in the GitHub repository.

Data note: the dataset is public (UCI, CC BY 4.0). Counts, shares and the p-chart in this report were cross-checked against the clean CSV; test statistics (ANOVA, Games-Howell, chi-square) are as reported by jamovi.